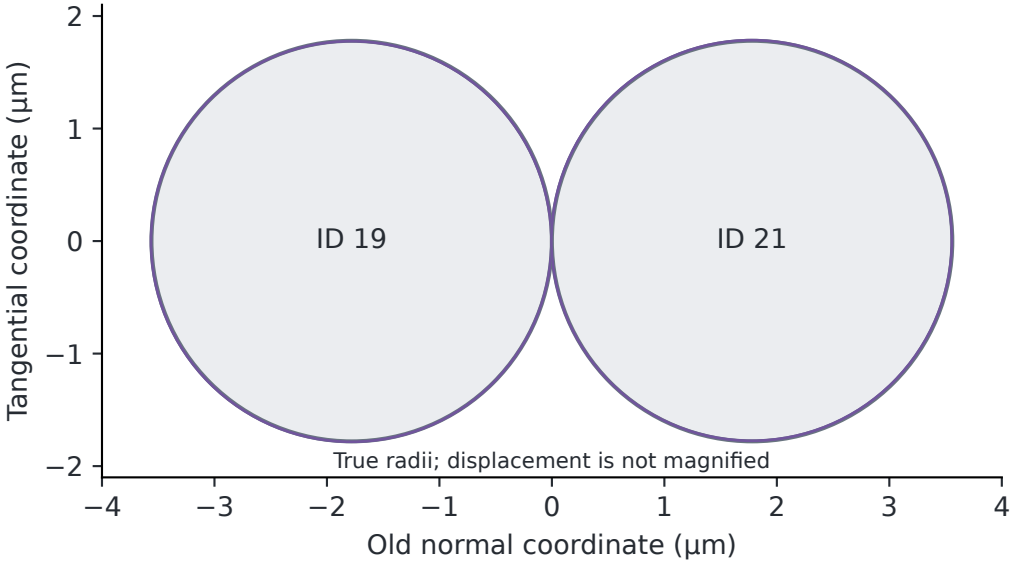
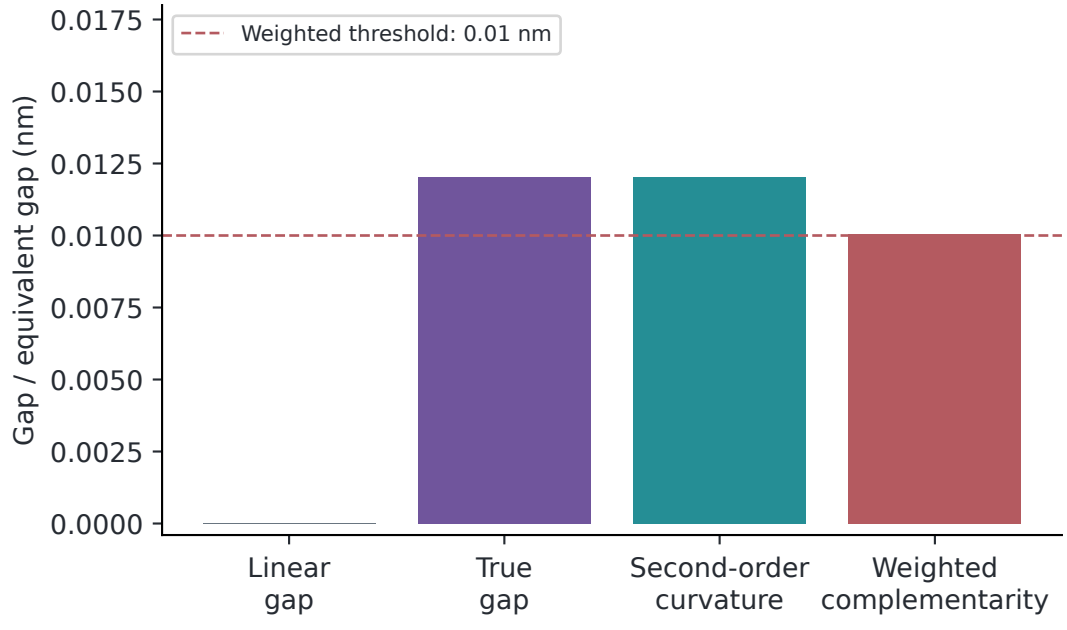
F10Q | Finite-sphere geometry and contact consistency

a Actual rejected pair at 0.464 ms



b Curvature quantitatively explains the gap

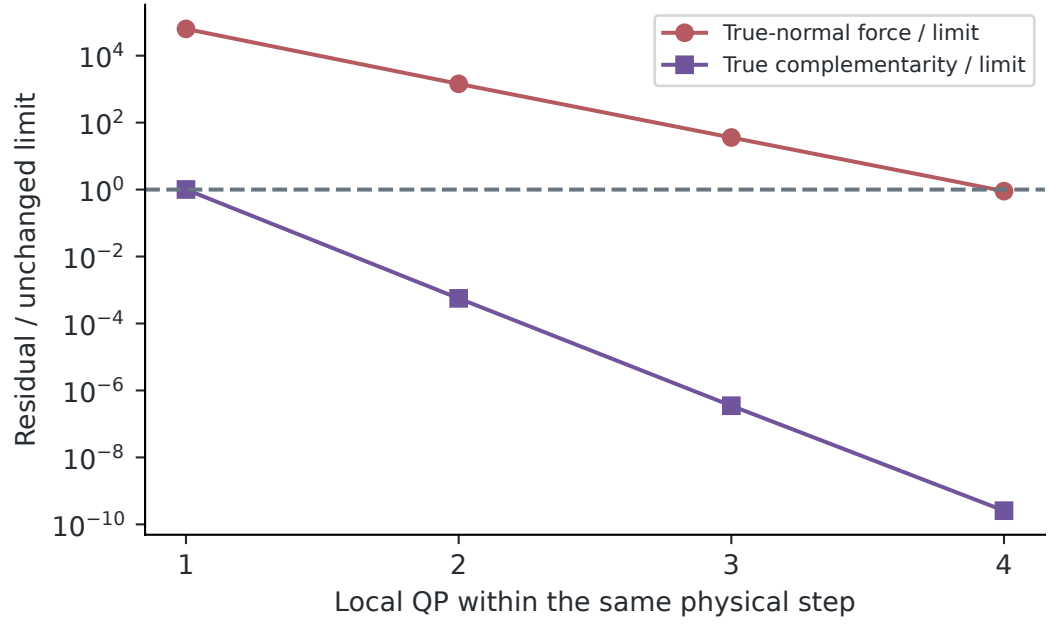


c Actual kinematics and reaction

Tangential relative displacement: 9.261203 nm
Normal relative displacement: -0.011488 nm
Normal rotation: 0.149052°
Pair reaction: 1.202162 nN
Normalization force: 1.439959 nN

Exact curvature identity error < 3.9e-12 nm
Pair mapping and SI-unit residual reproduced.
No new 2.1R graph edge at this rejected step.

d Frozen proposal: all gates pass at QP 4



Actual F10P rejected proposal and F10Q frozen-state replay. Frozen F/A; repeated QPs do not advance physical time.
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